Questions about Groups, Asked of Cayley Graphs

Ramsey Properties, Hamilton Cycles, Diameter and Product-Free Sets

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Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. Question 4: Product-Free Sets
6. The Translation
7. Progress
8. Open Problems

Recap: Cayley Graphs

Definition 1 (deck 00)

For a group G and $S = S^{-1} \subseteq G \setminus \{e\}$, the **Cayley graph** $\Gamma(G, S)$ has vertex set G and edges $\{g, gs\}$, $s \in S$.

Connected $\iff$ S generates G; vertex-transitive ($x \mapsto hx$ is an automorphism). Deck 02 coloured the arc $g \to gs$ by s.

Four Questions

Four questions about a group — and the graph that answers each.

about the group	about the graph
explicit Ramsey graphs from groups?	clique number of $\Gamma(G, S)$
can G be listed with steps in S?	Hamilton cycle in $\Gamma(G, S)$
how long are words in S?	diameter of $\Gamma(G, S)$
large product-free sets?	independent sets in $\Gamma(G, X)$

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. Question 4: Product-Free Sets
6. The Translation
7. Progress
8. Open Problems

Motivation: Explicit Ramsey Graphs

▷ Erdős–Szekeres (1935): every graph on N vertices has a clique or an independent set of size $\geq \frac{1}{2}\log_2 N$ (Schildkraut, §1.1).

▷ Erdős (1947): almost every graph has neither of size $> 2\log_2 N$ (Conlon–Fox–Pham–Yepremyan, §1).

▷ A graph is **C-Ramsey** if both are $\leq C\log N$. "No explicit constructions of C-Ramsey graphs are known for any constant C" (Schildkraut, §1.1).

Random graphs work, but nobody can write one down.

Why Groups? The Paley Graph

Definition 2 (see CFPY, §1)

For a prime $N \equiv 1 \pmod 4$, the **Paley graph** P_N is the Cayley graph on $\mathbb{Z}_N$ whose generating set is the non-zero squares.

▷ "It has also long been conjectured that the family of Paley graphs should be a good source of Ramsey graphs" (CFPY, §1).

▷ Cayley graphs are structured, so one can hope to **prove** things about them.

Example 3 (checked)

P_{13}, P_{17}: clique number 3, independence number 3.

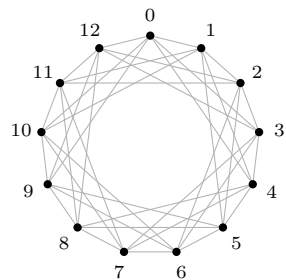


Figure: P_{13}: $x \sim y$ iff $x - y \in \{\pm 1, \pm 3, \pm 4\}$.

Alon's Conjecture

Conjecture 1 (Alon, 1995, 2002)

There is a constant C such that every finite group has a C-Ramsey Cayley graph.

▷ Motivated "by applications to coding theory" (Schildkraut, §1.1); Alon–Orlitsky asked, from information theory, for self-complementary Ramsey Cayley graphs (CFPY, §1).

▷ Why not random generators? In $\mathbb{F}_2^n$, if all of a subgroup H enters S, then H is a clique; so random Cayley graphs on $\mathbb{F}_2^n$ have clique number $\Theta(\log N \log\log N)$ (Green) (CFPY, §1.1).

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. Question 4: Product-Free Sets
6. The Translation
7. Progress
8. Open Problems

Motivation: Can a Group Be Walked Through?

▷ Gray codes: list the binary strings of length n so that consecutive ones differ in one bit: $000, 001, 011, 010, 110, 111, 101, 100$ — a Hamilton cycle in $\Gamma(\mathbb{Z}_2^n, \{e_1, \ldots, e_n\})$.

▷ Strasser (1959) conjectured that every connected Cayley graph is Hamiltonian (Draganić et al., Conj. 1.7).

▷ Lovász (1969) *asked* for a connected vertex-transitive graph without a Hamilton path; the conjecture is traditionally stated positively (Christofides–Hladký–Máthé, §1).

▷ Only five connected vertex-transitive graphs without a Hamilton cycle are known: K_2, Petersen, Coxeter, and the two obtained from them by replacing each vertex by a triangle; none but K_2 is a Cayley graph (CHM, §1).

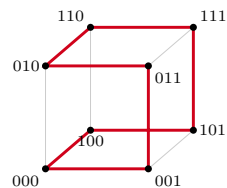


Figure: A Gray code as a Hamilton cycle (checked).

The Conjectures

Conjecture 2 (Lovász, 1969; see Ma–Zhao, §1)

Every connected vertex-transitive graph has a Hamilton path.

Conjecture 3 (Strasser, 1959; see Draganić et al., Conj. 1.7)

Every connected Cayley graph on at least 3 vertices has a Hamilton cycle.

▷ A stronger conjecture for general vertex-transitive graphs asks for Hamilton cycles apart from the known small exceptions (Conj. 1.6).

▷ The Cayley conjecture holds for abelian groups (§1.1).

▷ Babai conjectured infinitely many Cayley graphs with no cycle longer than $(1-c)n$ (CHM, §1). The new 2026 bounds sharply constrain this competing possibility.

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. **Question 3: How Long Are Words?**
5. Question 4: Product-Free Sets
6. The Translation
7. Progress
8. Open Problems

Motivation: Diameters of Groups

Definition 4 (Helfgott–Seress, §1)

For generators A of G, $\operatorname{diam}\Gamma(G, A)$ is the least ℓ such that every $g \in G$ is a word of length $\leq \ell$ in $A \cup A^{-1}$. $\operatorname{diam}(G) = \max_A \operatorname{diam}\Gamma(G, A)$.

▷ A long-standing problem in computer science: algorithms on permutation groups need short words; "on a more playful level, bounds on the diameter of permutation groups are relevant to every permutation puzzle (e.g., Rubik's cube)" (Helfgott–Seress, §1).

▷ Expansion: Cayley graphs of $SL_2(\mathbb{F}_p)$ give expander families (Bourgain–Gamburd), with applications in number theory (Green's survey).

Example 5 (S_n, generators $(12), (12\cdots n)^{\pm 1}$; checked)

Diameter $2, 6, 10, 15, 21$ for $n = 3, \ldots, 7$ — polynomial in n, while $|S_n| = n!$.

Babai's Conjecture

Conjecture 4 (Babai; published in Babai–Seress, 1992, Conj. 1.7; see Helfgott–Seress)

For every finite nonabelian simple group G, $\operatorname{diam}(G) \leq (\log |G|)^{O(1)}$.

▷ For $\mathrm{Alt}(n)$ this is the "folklore" conjecture $\operatorname{diam} = n^{O(1)}$ (Helfgott–Seress, §1).

▷ The nonabelian hypothesis is essential: even $\mathbb{Z}_p$ with $A = \{1\}$ has diameter $\lfloor p/2 \rfloor$, far from $(\log p)^{O(1)}$.

▷ The tool: **growth**. If $|AAA| \geq |A|^{1+\delta}$ for every generating A that is not almost all of G, then $O(\log \log |G|)$ triplings fill G.

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. **Question 4: Product-Free Sets**
6. The Translation
7. Progress
8. Open Problems

Motivation: Sum-Free Sets Without Commutativity

▷ Erdős: every n integers contain a sum-free subset of size $n/3$ (dilate into the middle third mod p); the same in abelian groups (Gowers, §1).

▷ Babai–Sós (1985): a non-trivial coset of an index-k subgroup is product-free; via the classification, every group of order n has one of size $cn^{4/7}$; Kedlaya (1997): $cn^{11/14}$ (Gowers, §1).

Problem 6 (Babai–Sós, 1985; see Gowers)

Does every finite group G have a product-free set X (no $x, y, z \in X$ with $xy = z$) of size $c|G|$?

Example 7 (checked)

In $\mathbb{Z}_7$: $\{3, 4\}$ is sum-free, and no sum-free set has 3 elements.

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. Question 4: Product-Free Sets
6. The Translation
7. Progress
8. Open Problems

Proposition 1: Cliques Are Difference Sets

Proposition 8

$X \subseteq G$ is a clique of $\Gamma(G, S)$ $\iff$ $X^{-1}X \setminus \{e\} \subseteq S$, and independent $\iff$ $X^{-1}X \cap S = \emptyset$.

Proof: $x \sim y \iff x^{-1}y \in S$ (equivalently $y^{-1}x \in S$, as $S = S^{-1}$). CFPY use edges $\{x, y\}$ with $xy^{-1} \in S$; then $X^{-1}X$ becomes XX^{-1}.

Fact 9 (Conlon–Fox–Pham–Yepremyan, §1)

*Colour the edge xy of K_N (vertices G) by $\{xy^{-1}, yx^{-1}\}$. Every colour class has maximum degree ≤ 2, and $\Gamma(G, S)$ is the union of the colour classes in S: an **entangled graph**.*

▷ Counting cliques of a random Cayley graph = counting vertex sets that span **few colours**: sets A with $|AA^{-1}| \leq K|A|$ (CFPY, §1.1).

▷ CFPY prove the count for **every** bounded-degree colouring of K_N: "Our proof does not make use of the underlying group structure" (abstract).

Proposition 2: Hamilton Cycles Are Words

Proposition 10

$\Gamma(G, S)$ has a Hamilton cycle $\iff$ there are $s_1, \ldots, s_n \in S$ ($n = |G|$, repetitions allowed) with $s_1 s_2 \cdots s_n = e$ and the partial products $e, s_1, s_1 s_2, \ldots, s_1 \cdots s_{n-1}$ all distinct.

Proof: the cycle $g_1 g_2 \cdots g_n$ translated to start at e (left multiplication is an automorphism); $s_i = g_i^{-1} g_{i+1}$.

▷ Compare deck 02: an **R-sequencing** is the case $S = G \setminus \{e\}$ with every s used **exactly once** (a rainbow cycle); here letters may repeat.

▷ Gray code: $S = \{e_1, e_2, e_3\}$, word $e_3 e_2 e_3 e_1 e_3 e_2 e_3 e_1$.

Proposition 3: Diameter Is Growth

Proposition 11

Let $B = A \cup A^{-1} \cup \{e\}$. *Then* $\operatorname{diam} \Gamma(G, A) \leq \ell \iff B^\ell = G$. *The ball of radius* k *around* e *is* B^k.

Proof: the vertices at distance $\leq k$ from e are the products of at most k letters from $A \cup A^{-1}$.

Theorem 12 (Helfgott, 2008, abstract)

"Every subset of $SL_2(\mathbb{Z}/p\mathbb{Z})$ *grows rapidly when it acts on itself by the group operation. It follows readily that, for every set of generators* A *..., every element ... can be expressed as a product of at most* $O((\log p)^c)$ *elements of* $A \cup A^{-1}$*."*

Growth means $|AAA| \geq |A|^{1+\delta}$ unless $AAA = G$ (Pyber–Szabó, abstract). Iterating, $|A^{3^k}| \geq |A|^{(1+\delta)^k}$, so $O(\log \log |G|)$ triplings reach G, and the radius grows like $3^k = (\log |G|)^{O(1)}$.

Proposition 4: Product-Free Sets Are Independent

Proposition 13

*X is product-free $\iff$ in the Cayley digraph $\mathrm{Cay}(G, X)$ (arcs $g \to gx$, $x \in X$) the set X spans **no arc**.*

Proof: an arc inside X is $x \to xy$ with $x, xy \in X$, $y \in X$: a solution of $xy = z$.

$\triangleright$ Gowers works with the **bipartite** Cayley graph and shows it is **quasirandom** (edges spread evenly) when G has no small non-trivial representation; quasirandomness forces $ab = c$ for any three large sets (Gowers, §1).

The Dictionary

group	**graph**
$X^{-1}X \setminus \{e\} \subseteq S$	X is a clique of $\Gamma(G, S)$
sets with $\lvert AA^{-1}\rvert \leq K\lvert A\rvert$	vertex sets spanning few colours
word $s_1 \cdots s_n = e$, distinct partial products	Hamilton cycle
$B^\ell = G$	diameter $\leq \ell$
growth $\lvert A^3\rvert \geq \lvert A\rvert^{1+\delta}$	expansion of balls
product-free X	X spans no arc of $\mathrm{Cay}(G, X)$
$\mathrm{Aut}\,\Gamma(G, S) = G$	GRR (Godsil; Xia–Zheng)

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. Question 4: Product-Free Sets
6. The Translation
7. **Progress**
8. Open Problems

Ramsey Cayley Graphs: The Current Baseline

▷ (Alon) random Cayley graphs: clique number $O(\log^2 N)$.

▷ (Green; Green–Morris) $\mathbb{Z}_N$: $C \log N$, and $(2 + o(1)) \log N$ for N prime.

▷ (Conlon–Fox–Pham–Yepremyan, 2024) every group: $O(\log N \log\log N)$; Alon's conjecture for abelian groups of almost all orders; self-complementary $(2 + o(1))$-Ramsey Cayley graphs over $\mathbb{F}_q^n$, $q \equiv 1 \pmod 4$.

▷ (Schildkraut, December 2025) Alon's conjecture for all groups of almost all orders.

Next we vary the density of the random generator set, then examine the sparse regime of the 2026 colouring theorem.

Theorem 14 (Alon–Pham, September 2025, Thm. 4)

For an abelian group of order n and $p \leq 1/2$, random Cayley graphs in the binomial symmetric-generator model, and random Cayley sum graphs, have independence number at most $\widetilde{O}(p^{-3/2})$ with high probability.

In a Cayley sum graph, xy is an edge when $x+y$ belongs to the sampled set. Thus an independent set A has all sums $x+y$ with distinct $x, y \in A$ outside that set.

▷ The algebraic question about avoiding sums becomes an independence question.

▷ The source also bounds complements of large nonsumsets (Thm. 5).

$\widetilde{O}$ suppresses logarithmic factors in n. This is 2025 background for the sparse random regime, not a new 2026 theorem.

Sparse Random Cayley Graphs Can Be 3-Coloured

Theorem 15 (Tung, June 2026; revised August, Thm. 1.1)

Let G be any abelian group of order N. Sample $\lfloor \ln N/64 \rfloor$ independent uniform elements of G and form the undirected Cayley graph (discarding loops). With probability tending to 1 as $N \to \infty$, its chromatic number is at most 3.

The number of sampled elements is logarithmic in N; sampling is with replacement, and the graph uses the symmetrised nonzero set. The sampled elements need not generate G.

To interpret this algebraically, ask how to partition G so that no part contains a difference from the sampled generator set.

Three Colours Mean Three Difference-Avoiding Sets

Let S be the symmetrised nonzero generator set in the preceding theorem. A proper 3-colouring is exactly a partition

$$G = A_1 \sqcup A_2 \sqcup A_3, \qquad (A_i - A_i) \cap S = \varnothing \quad (i = 1, 2, 3).$$

▷ Two vertices of A_i are adjacent exactly when their difference lies in S.

▷ Thus each part is independent, and some part has size at least $N/3$.

▷ Tung's 2026 theorem guarantees such a partition with high probability in this sparse sampling model.

Alon's Ramsey question seeks small independent sets. Here the graph has a large independent set because the generator set is sparse.

Hamilton Cycles Below Linear Degree

▷ (Christofides–Hladký–Máthé, 2014) for each $\varepsilon > 0$, sufficiently large connected vertex-transitive graphs with degree at least εn are Hamiltonian.

Theorem 16 (Bedert–Draganić–Müyesser–Pavez-Signé, 2026)

For some absolute $c \geq 1/200$, every sufficiently large connected n-vertex Cayley graph of degree $d > n^{1-c}$ has a Hamilton cycle. In particular, $d > n^{199/200}$ suffices.

Translation: for every corresponding generating set $S = S^{-1}$, there is a word of length n in S whose partial products visit G exactly once before returning to e.

The proof uses weak arithmetic regularity from the Graham rearrangement programme in deck 02; only the conclusion is used here.

Hamilton Cycles: Four Distinct Prime Factors

Theorem 17 (Lehner–Maghsoudi–Miraftab, July 2026, Thm. 1.3)

If $|G| = pqrs$, where p, q, r, s are four distinct primes, every connected Cayley graph on G has a Hamilton cycle.

▷ The primes need not all be odd: the theorem includes even order.

▷ Morris (2021) proved the case of four distinct **odd** primes. The earlier result should not be stated for unrestricted $pqrs$.

▷ A different direction: $C \log n$ random generators give a Hamiltonian Cayley graph with high probability (Draganić et al., 2024).

These are exact Hamiltonicity results under extra assumptions on the group order or the generator set.

Without Extra Hypotheses: How Much of the Group Can We Visit?

The preceding results give a Hamilton cycle under density or group-order assumptions. For an arbitrary connected Cayley graph, relax the target.

The same word problem, with length $3 \le k \le |G|$

Find $s_1, \dots, s_k \in S$ with $s_1 \cdots s_k = e$ such that $e, s_1, s_1 s_2, \dots, s_1 \cdots s_{k-1}$ are distinct.

- $\triangleright$ $k = |G|$: a Hamilton cycle.
- $\triangleright$ $k \ge c|G|$: a cycle visiting a fixed positive fraction of the group.
- $\triangleright$ $k \ge (1 - o(1))|G|$: a cycle missing a vanishing fraction of the group.

The 2026 results below apply to the larger class of connected vertex-transitive graphs, hence also to Cayley graphs (Ma–Zhao, §1).

Long Cycles: The 2026 Timeline

Let $\mathrm{circ}(\Gamma)$ be the maximum cycle length. Below, Γ is any connected vertex-transitive graph on sufficiently many vertices.

▷ June (revised July): for each $\varepsilon > 0$, $\mathrm{circ}(\Gamma) \geq n^{2/3-\varepsilon}$ (Bucić–Christoph–Pokrovskiy–Steiner).

▷ September: for each $\varepsilon > 0$, $\mathrm{circ}(\Gamma) \geq n^{1-\varepsilon}$ (Li–Methuku). A parallel BCPS working draft with the same bound is recorded in Ma–Zhao, §1.

▷ October: $\mathrm{circ}(\Gamma) \geq cn$ for an absolute $c > 0$ (Ma–Zhao).

An exponent approaching 1 need not give a positive fraction of n: $n^{1-o(1)}$ and $(1 - o(1))n$ are different guarantees.

Almost Hamilton Cycles: Degree Assumptions in 2026

Theorem 18 (Christoph–Hunter–Sudakov, 24 September 2026, Thm. 1.8)

Every connected d-regular vertex-transitive graph on n vertices with

$$\frac{d}{(\log n)^3 \log\log n} \longrightarrow \infty$$

has a cycle of length $(1 - o(1))n$ as $n \to \infty$.

▷ For Cayley graphs, this gives a closed word visiting all but $o(|G|)$ group elements, under the stated degree assumption.

▷ The condition requires the degree to grow faster than $(\log n)^3 \log\log n$.

▷ The next theorem first guarantees a linear cycle with no degree assumption, then improves the almost-spanning degree regime.

Ma–Zhao: A Positive Fraction at Every Degree

Theorem 19 (Jie Ma–Ziyuan Zhao, 1 October 2026, Thm. 1.1)

There is an absolute constant $c > 0$ such that every finite connected vertex-transitive graph Γ on $n \geq 3$ vertices satisfies

$$\operatorname{circ}(\Gamma) \geq cn.$$

▷ The same constant works for all these graphs, regardless of degree.

▷ This applies to every connected Cayley graph on at least three vertices.

▷ It strengthens a bound $n^{1-\varepsilon}$ to a fixed positive fraction: $n^{1-\varepsilon}/n = n^{-\varepsilon} \to 0$ for fixed $\varepsilon > 0$.

Ma–Zhao: Almost All Vertices When Degree Grows

Theorem 20 (Ma–Zhao, Thm. 1.1, degree-dependent conclusion)

There is an absolute integer d_0 such that every connected vertex-transitive graph on $n \geq 3$ vertices and of degree $d \geq d_0$ has

$$\operatorname{circ}(\Gamma) \geq (1 - d^{-1/100})n.$$

▷ As $d \to \infty$, the fraction of uncovered vertices tends to zero. There is no required growth rate relative to n.

▷ This improves the degree condition in the September result of Christoph–Hunter–Sudakov.

▷ The estimate allows uncovered vertices. Exact Hamiltonicity remains a stronger demand.

Ma–Zhao in the Language of Group Words

Let $|G| \geq 3$ and let $S = S^{-1} \subseteq G \setminus \{e\}$ generate G. A cycle in $\mathrm{Cay}(G, S)$ can be translated to start at e. Its edge labels give

$$s_1 \cdots s_k = e, \qquad e, s_1, s_1 s_2, \ldots, s_1 \cdots s_{k-1} \text{ pairwise distinct.}$$

Consequence of Ma–Zhao, Thm. 1.1

Such a word exists with $k \geq c|G|$; for $|S| = d \geq d_0$, it exists with $k \geq (1 - d^{-1/100})|G|$.

The generators in the word may repeat. The requirement is on distinct visited group elements; deck 02 additionally prescribes distinct edge colours.

If a family has no cycle longer than $(1 - \delta)|G|$ for fixed $\delta > 0$, its degrees must be bounded in terms of δ (a direct consequence).

Directed Cayley Graphs: Keep the Distinction

Without $S = S^{-1}$, only the prescribed directed steps are available. The broader class is connected vertex-transitive digraphs.

▷ February: every such n-vertex digraph has a directed cycle of length $\Omega(n^{1/3})$ (Bucić–Hendrey–Mohar–Steiner–Yepremyan).

▷ July (revised August): the lower bound improves to $\Omega(\sqrt{n})$ (Li–Methuku, Thm. 1.5).

▷ The same paper gives infinitely many vertex-transitive digraphs with every directed cycle missing at least $n/12$ vertices (Thm. 1.3).

The linear-gap examples are stated for **vertex-transitive digraphs**; no Cayley property is asserted here. They do not refute the undirected Cayley Hamilton-cycle conjecture.

Diameter: The Classical Baseline

▷ (Helfgott, 2008) polylogarithmic diameter for $SL_2(\mathbb{F}_p)$.

▷ (Pyber–Szabó), (Breuillard–Green–Tao): finite simple groups of Lie type of **bounded rank**.

▷ (Helfgott–Seress, 2014) $\operatorname{diam}(\mathrm{Sym}(n)) \leq \exp(O((\log n)^4 \log\log n))$.

Babai's polylogarithmic conjecture concerns **nonabelian simple groups**. Even $\mathbb{Z}_p$ with generators $\{\pm 1\}$ has diameter $\lfloor p/2 \rfloor$. To treat arbitrary finite groups, new bounds must reflect their algebraic structure.

Returning to Word Length: Solubility and Exponent

A long simple cycle controls how many elements one walk can visit. Diameter asks how quickly a shortest word can reach each element.
A group is **soluble** if its iterated commutator subgroups eventually become trivial. Its **exponent** e is the least positive integer such that $g^e = 1$ for every $g \in G$.

Example 21 (two abelian, hence soluble, groups)

$\mathbb{Z}_m$ has exponent m. The group $(\mathbb{Z}_2)^r$ has exponent 2 and order 2^r.

Exponent and group order can therefore be very different. The 2026 bound keeps track of both (Eberhard et al., Thm. 1.2).

Diameter in 2026: Soluble Groups

Let G be a finite soluble group of exponent e.

Theorem 22 (Eberhard–Maini–Sabatini–Tracey, April 2026, Thm. 1.2)

Every finite soluble group G of exponent e satisfies

$$\operatorname{diam}(G) \ll e(\log|G|)^8.$$

The diameter is maximised over generating sets. Equivalently, for every generating A, every group element has a word of length at most $Ce(\log|G|)^8$ in $A \cup A^{-1}$, for an absolute C.

The exponent factor records an obstruction already visible in cyclic groups.

Transitive Soluble Permutation Groups

A subgroup $G \leq \mathrm{Sym}(n)$ is **transitive** if it has one orbit on the n points. Here n is the permutation degree, not $|G|$.

Theorem 23 (Eberhard–Maini–Sabatini–Tracey, Thm. 1.7)

Every transitive soluble permutation group $G \leq \mathrm{Sym}(n)$ *satisfies* $\mathrm{diam}(G) = O(n^5)$.

Example 24 (permuting three positions)

S_3 acts transitively on 3 points, so $n = 3$, while $|S_3| = 6$. Its Cayley graph has 6 vertices, one for each permutation. The group is soluble: its commutator subgroup is the cyclic group A_3.

The theorem bounds word length using the number of positions being permuted, rather than the number of vertices in the Cayley graph.

How Babai's Conjecture Would Control All Transitive Actions

Now drop the solubility assumption. Let $G \leq \mathrm{Sym}(n)$ be transitive.

Conditional consequence (Cor. 1.5)

If Babai's diameter conjecture holds for nonabelian finite simple groups, then every transitive permutation group of degree n has diameter $n^{O(1)}$.

Thus a diameter statement for simple groups controls short-word questions for all transitive permutation groups. This conclusion remains conditional.

Product-Free Sets and Regular Representations

▷ (Gowers, 2008) $PSL_2(q)$ has no product-free set of size $Cn^{8/9}$, where $n = |PSL_2(q)|$: quasirandomness answers the large-set question.

▷ Godsil classified which groups have a graphical regular representation (GRR).

▷ (Morris–Spiga, 2018) almost all Cayley digraphs are directed regular representations.

▷ (Xia–Zheng) almost all Cayley graphs are GRRs: the Babai–Godsil–Imrich–Lovász conjecture.

A regular representation has no automorphisms beyond the regular group action.

Contents

1. Recap
2. Question 1: Ramsey Graphs from Groups
3. Question 2: Listing a Group by Generators
4. Question 3: How Long Are Words?
5. Question 4: Product-Free Sets
6. The Translation
7. Progress
8. Open Problems

Open Problem 1: Alon's Conjecture

Conjecture 5 (Alon)

There is C such that every finite group G has $S = S^{-1} \subseteq G \setminus \{e\}$ for which no $X \subseteq G$ with $|X| > C\log|G|$ has $X^{-1}X \setminus \{e\} \subseteq S$ or $X^{-1}X \cap S = \emptyset$.



Proposition 8

Graph form

Every finite group G has a Cayley graph $\Gamma(G, S)$ with **clique number and independence number at most** $C\log|G|$.

Known. $\mathbb{Z}_N$ (Green; Green–Morris); abelian groups of almost all orders (CFPY); all groups of almost all orders (Schildkraut).
Open: the remaining orders — and an explicit construction.

Open Problem 2: Hamilton Cycles in Cayley Graphs

Conjecture 6 (Strasser, 1959)

For every finite group G, $|G| \geq 3$, and generating $S = S^{-1}$, there are $s_1, \ldots, s_n \in S$ with $s_1 \cdots s_n = e$ and distinct partial products $e, s_1, \ldots, s_1 \cdots s_{n-1}$.



Proposition 10

Graph form

Every connected Cayley graph on at least 3 vertices has a **Hamilton cycle**.

Known. abelian groups; $d > n^{199/200}$ for large n (2026); order $pqrs$ for four distinct primes (2026); random generators (2024). Linear circumference (Ma–Zhao, 2026) does not yet give a Hamilton cycle.

Open Problem 3: Babai's Conjecture

Conjecture 7 (Babai; Babai–Seress, 1992)

There is C such that for every finite nonabelian simple group G and every generating set A, every element of G is a product of at most $(\log|G|)^C$ elements of $A \cup A^{-1}$.



Proposition 11

Graph form

Every connected Cayley graph of a finite nonabelian simple group G has **diameter at most** $(\log|G|)^C$.

Known. bounded-rank groups of Lie type (Pyber–Szabó), (BGT); $\mathrm{Alt}(n)$: quasipolynomial (Helfgott–Seress). **Open:** unbounded rank, and the polynomial bound $n^{O(1)}$ for $\mathrm{Alt}(n)$.

Open Problem 4: Long Cycles, the Other Way

Conjecture 8 (Babai; see Christofides–Hladký–Máthé, §1)

There are $c > 0$ and infinitely many connected Cayley graphs Γ with no cycle longer than $(1-c)|\Gamma|$.

$\Updownarrow$ Proposition 10, for cycles of length k

Word form

There are infinitely many groups G with generating sets $S = S^{-1}$ such that every word $s_1 \cdots s_k = e$ in S with distinct partial products has $k \leq (1-c)|G|$.

Known. every connected vertex-transitive graph on $n \geq 3$ vertices has a cycle of length at least cn (Ma–Zhao, 2026). Their degree-dependent bound forces any proposed fixed-gap family to have bounded degree. Open Problems 2 and 4 cannot both be true.

Summary

▷ **Ramsey**: groups as candidates for explicit Ramsey graphs; cliques are sets with few "differences" $X^{-1}X$ — counted in any proper-ish colouring.

▷ **Hamiltonicity**: listing a group by generators; open since 1959.

▷ **Diameter**: short words = small diameter = growth of products; Babai's conjecture.

▷ **Product-free sets**: independence in the Cayley digraph; answered by quasirandomness (Gowers).

One graph per group — and a different question for each property.

Thanks !

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